

Self-Assessment Test Report — (Optimized Implementation)

Environment notice: these figures were collected on an environment compliant with guideline section 2.1 (Linux + GNU GCC + x86-64) and may be used as the official self-assessment result.

Auto-generated by `gen_report.py` from `reports/*.json`, following guideline section 3.5.

1. Algorithm Information

Item	Content
Category	Public-key cryptography
Function	Digital signature
Name	DOVE (classic / pkc+skc)
Implementation version	Optimized Implementation
Parameter sets / security levels	DOVE_classic_128 / DOVE_classic_256 / DOVE_classic_512 / DOVE_pkc_skc_128 / DOVE_pkc_skc_256 / DOVE_pkc_skc_512 (claimed 128 / 256 / 512 / 128 / 256 / 512-bit security)
Variants	classic (full PK) / pkc_skc (compact PK+SK)
Programming interface	<code>sig_keygen</code> / <code>sig_sign</code> / <code>sig_verify</code>
Underlying hash	SM3

2. Evaluation Environment

Item	This run (compliant)	Compliance requirement (section 2.1)
Processor	x86_64 (Linux)	64-bit x86, x86-ISA compatible, >2GHz recommended

Item	This run (compliant)	Compliance requirement (section 2.1)
Cores	single process / single thread	single core only
Memory	15 GB	>4GB recommended
Operating system	Linux 6.18.33.2-microsoft-standard-WSL2 [OK]	Linux, kernel >= 4.19
Compiler	gcc (Ubuntu 9.4.0-1ubuntu1~20.04.2) 9.4.0 [OK]	GNU GCC >= 8.5.0
Build tool	CMake 3.16.3 [OK]	CMake >= 3.11.4
Instruction set	x86-64 base (RDTSC timing), AVX2 available	x86-64 + AVX2
Start / end time	2026-06-30 23:16:54 ~ 23:22:04 (local)	fill in as measured

3. Functional Test

Method: for each security level, a key pair is generated from a fixed DRNG seed; a fixed 64-byte message is signed and verified; a tampered signature must be rejected. Full KAT vectors are in `API_PKC/Test_Vector/` and `data/kat/`.

Security level	keygen/sign/verify correctness	forged-signature rejection	Result
DOVE_classic_128	PASS	PASS	PASS
DOVE_classic_256	PASS	PASS	PASS
DOVE_classic_512	PASS	PASS	PASS
DOVE_pkc_skc_128	PASS	PASS	PASS
DOVE_pkc_skc_256	PASS	PASS	PASS
DOVE_pkc_skc_512	PASS	PASS	PASS

4. Performance Test (CPU Cycles)

Each operation is run **100 times**; signing uses a fixed 64-byte message. Average CPU cycles (RDTSC) and throughput (ops/s) are reported. KAT generation is not included.

Parameter set	Operation	Avg CPU cycles	ops/s
DOVE_classic_128	KeyGen	42,329,477	68.94
DOVE_classic_128	Sign	2,764,061	1055.75
DOVE_classic_128	Verify	5,781,061	504.79
DOVE_classic_256	KeyGen	457,651,534	6.38
DOVE_classic_256	Sign	21,295,644	137.04
DOVE_classic_256	Verify	28,217,290	103.42
DOVE_classic_512	KeyGen	6,760,373,809	0.43
DOVE_classic_512	Sign	250,848,590	11.63
DOVE_classic_512	Verify	421,307,321	6.93
DOVE_pkc_skc_128	KeyGen	28,994,237	100.65
DOVE_pkc_skc_128	Sign	14,959,585	195.08
DOVE_pkc_skc_128	Verify	13,260,626	220.07
DOVE_pkc_skc_256	KeyGen	281,021,579	10.38
DOVE_pkc_skc_256	Sign	135,285,378	21.57
DOVE_pkc_skc_256	Verify	112,308,877	25.99
DOVE_pkc_skc_512	KeyGen	4,858,325,659	0.60
DOVE_pkc_skc_512	Sign	1,532,800,858	1.90
DOVE_pkc_skc_512	Verify	1,358,369,362	2.15

5. Key and Signature Sizes

Parameter set	Public key (B)	Private key (B)	Signature (B)
DOVE_classic_128	191,646	192,990	136

Parameter set	Public key (B)	Private key (B)	Signature (B)
DOVE_classic_256	2,050,352	1,925,976	296
DOVE_classic_512	22,833,052	20,181,508	648
DOVE_pkc_skc_128	43,576	24	136
DOVE_pkc_skc_256	446,992	40	296
DOVE_pkc_skc_512	5,062,192	72	648

6. Raw Evidence Index

Item	Content
Test command	<code>run_self_assessment.sh 100</code>
Input data	fixed seed + 64B message (no KAT generation)
Raw log	<code>logs/self_assessment.log</code>
Structured results	<code>reports/<algorithm>_<timestamp>.json</code> , <code>reports/summary.csv</code>

- Benchmark code: `bench/sig_bench.c` , adapter: `adapter/sig/` , entry: `src/main.c`
- Reproduction: run `run_self_assessment.sh` on a compliant host.